TP 1 : Base de données

Modèle relationnel

student(*idstudent*, *firstname*, *lastname*, *field*, *average*, *absence*, *city*)

idstudent	firstname	lastname	field	average	absence	city
1	Ali	Hassan	Mathematics	17.5	2	Rabat
2	Sarah	Mohammad	Economics	18.3	0	Rabat
3	Omar	Yussuf	Mathematics	16.0	0	Kénitra
4	Laila	Ahmad	Economics	14.4	0	Rabat
5	Khalid	Ali	Mathematics	19.2	0	Salé
6	Fatima	Salem	Economics	15.7	10	Kénitra
7	Noor	Hussein	Mathematics	18.1	0	Salé
8	Ibrahim	Zaid	Economics	19.8	0	Salé
9	Aya	Nasser	Mathematics	14.5	2	Salé
10	Huda	Fahim	Economics	13.0	3	Rabat
11	Salma	Jamal	Mathematics	17.3	0	Salé
12	Yassin	Othman	Economics	12.4	1	Rabat
13	Amina	Sami	Mathematics	19.2	1	Rabat
14	Hassan	Rami	Economics	18.7	0	Rabat
15	Tariq	Bassam	Mathematics	15.9	10	Kénitra
16	Lina	Aref	Economics	14.2	0	Rabat
17	Zainab	Hadi	Mathematics	17.5	5	Rabat
18	Rashid	Kamal	Economics	19.1	0	Kénitra
19	Aisha	Fadel	Mathematics	12.6	0	Rabat
20	Noor	Mahmood	Economics	16.7	2	Rabat

Table 1 – Extrait de la relation student

1. Créer la relation *student* en précisant les contraintes d'intégrité.
2. Insérer les données.
3. Mettre à jour la table dans les deux cas suivants :
   - 3.1 L'étudiante *Fatima Salem*, dont l'identifiant est 6, a justifié 5 heures d'absence.
   - 3.2 L'étudiant *Khalid Ali*, dont l'identifiant est 5, a quitté le centre.
4. Afficher :
   - 4.1 La liste des étudiants.
   - 4.2 Les villes des étudiants du centre.
   - 4.3 Les étudiants qui n'étaient jamais absents.
   - 4.4 Les prénoms et les noms des étudiants qui habitent à *Rabat*.
   - 4.5 Les moyennes des étudiants qui n'habitent pas à *Rabat*.
   - 4.6 La liste des étudiants de la filière *Mathematics* triée selon la moyenne en ordre décroissant.
   - 4.7 Les trois premiers étudiants de la filière *Economics*.
5. Afficher :
   - 5.1 La note moyenne des étudiants du centre.
   - 5.2 La note moyenne de chaque filière.
   - 5.3 La plus grande et la plus petite moyenne de la filière *Mathematics*.
   - 5.4 La somme des absences en *Economics*.
   - 5.5 Le nombre des étudiants du centre.
   - 5.6 Le nombre des étudiants par ville.
   - 5.7 Les villes dont le nombre des étudiants dépasse 10.

Corrigé

Question 1 : création de la relation

```sql
CREATE TABLE IF NOT EXISTS student (
    idstudent INTEGER PRIMARY KEY,
    firstname TEXT NOT NULL,
    lastname  TEXT NOT NULL,
    field     TEXT NOT NULL,
    average   REAL NOT NULL CHECK (average BETWEEN 0 AND 20),
    absence   INTEGER NOT NULL DEFAULT 0 CHECK (absence >= 0),
    city      TEXT NOT NULL
);
```

Question 2 : insertion des données

```sql
INSERT INTO student VALUES
(1, 'Ali', 'Hassan', 'Mathematics', 17.5, 2, 'Rabat'),
(2, 'Sarah', 'Mohammad', 'Economics', 18.3, 0, 'Rabat'),
(3, 'Omar', 'Yussuf', 'Mathematics', 16.0, 0, 'Kénitra'),
(4, 'Laila', 'Ahmad', 'Economics', 14.4, 0, 'Rabat'),
(5, 'Khalid', 'Ali', 'Mathematics', 19.2, 0, 'Salé'),
(6, 'Fatima', 'Salem', 'Economics', 15.7, 10, 'Kénitra'),
(7, 'Noor', 'Hussein', 'Mathematics', 18.1, 0, 'Salé'),
(8, 'Ibrahim', 'Zaid', 'Economics', 19.8, 0, 'Salé'),
(9, 'Aya', 'Nasser', 'Mathematics', 14.5, 2, 'Salé'),
(10, 'Huda', 'Fahim', 'Economics', 13.0, 3, 'Rabat'),
(11, 'Salma', 'Jamal', 'Mathematics', 17.3, 0, 'Salé'),
(12, 'Yassin', 'Othman', 'Economics', 12.4, 1, 'Rabat'),
(13, 'Amina', 'Sami', 'Mathematics', 19.2, 1, 'Rabat'),
(14, 'Hassan', 'Rami', 'Economics', 18.7, 0, 'Rabat'),
(15, 'Tariq', 'Bassam', 'Mathematics', 15.9, 10, 'Kénitra'),
(16, 'Lina', 'Aref', 'Economics', 14.2, 0, 'Rabat'),
(17, 'Zainab', 'Hadi', 'Mathematics', 17.5, 5, 'Rabat'),
(18, 'Rashid', 'Kamal', 'Economics', 19.1, 0, 'Kénitra'),
(19, 'Aisha', 'Fadel', 'Mathematics', 12.6, 0, 'Rabat'),
(20, 'Noor', 'Mahmood', 'Economics', 16.7, 2, 'Rabat');
```

Question 3 : mise à jour

Question 3.1

```sql
UPDATE student SET absence = absence - 5 WHERE idstudent = 6;
```

Question 3.2

```sql
DELETE FROM student WHERE idstudent = 5;
```

Question 4 : requêtes de sélection

Question 4.1

```sql
SELECT * FROM student;
```

Question 4.2

```sql
SELECT DISTINCT city FROM student;
```

Question 4.3

```sql
SELECT * FROM student WHERE absence = 0;
```

Question 4.4

```sql
SELECT firstname, lastname FROM student WHERE city = 'Rabat';
```

Question 4.5

```sql
SELECT average FROM student WHERE city != 'Rabat';
```

Question 4.6

```sql
SELECT * FROM student
WHERE field = 'Mathematics'
ORDER BY average DESC;
```

Question 4.7

```sql
SELECT * FROM student
WHERE field = 'Economics'
ORDER BY average DESC
LIMIT 3;
```

Question 5 : fonctions d'agrégation et regroupement

Question 5.1

```sql
SELECT AVG(average) AS Moyenne FROM student;
```

Question 5.2

```sql
SELECT field, AVG(average) AS Moyenne
FROM student
GROUP BY field;
```

Question 5.3

```sql
SELECT MIN(average) AS Note_min, MAX(average) AS Note_max
FROM student
WHERE field = 'Mathematics';
```

Question 5.4

```sql
SELECT SUM(absence) AS Total
FROM student
WHERE field = 'Economics';
```

Question 5.5

```sql
SELECT COUNT(idstudent) AS Total FROM student;
```

Question 5.6

```sql
SELECT city, COUNT(idstudent) AS nb_etudiants
FROM student
GROUP BY city;
```

Question 5.7

```sql
SELECT city, COUNT(idstudent) AS nb_students
FROM student
GROUP BY city
HAVING nb_students > 10;
```